

ARRAY QNO:5

QUESTION:

Problem Description:

For some reason, your school's football team has chosen to spell out the numbers on their jerseys instead of using the usual digits. Being great fans, you're going to be ready to cheer for your favorite players by bringing letter cards so you can spell out their number. Each fan has different favorites, so they each need to bring different sets of letters.

The English spellings for the numbers 0 to 12 are:

ZERO ONE TWO THREE FOUR FIVE SIX

SEVEN EIGHT NINE TEN ELEVEN TWELVE

Input Format:

Read a set of integers from 0 to 12, separated by spaces, representing one fan's favorite players. The last integer will be 999, marking the end of the line.

Output Format:

Print the same numbers, then a period and a space. Then, in alphabetical order, print all the letters the fan needs to be able to spell any one of the jersey numbers provided

Logical Test Cases

Test Case 1

INPUT (STDIN)

2 3 0999

Test Case 2

INPUT (STDIN)

8 14 0999

EXPECTED OUTPUT

2 3 0999. E E H O R T T W

EXPECTED OUTPUT

8 14 0999. E G H I T

Mandatory Test Cases

Test Case 1

KEYWORD

char nums[13][256]

Test Case 2

KEYWORD

for(n=0;n<26;n++)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

270

Test Case 3

NLOC

40

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char nums [13] [256] = {
        "ZERO", "ONE", "TWO", "THREE", "FOUR",
        "FIVE", "SIX", "SEVEN", "EIGHT", "NINE",
        "TEN", "ELEVEN", "TWELVE"
    };
    char input[100];
    int letter_counts[26] = {0};
    int first = 1;

    while (scanf("%s", input) == 1) {
        if (!first) {
            printf(" ");
        }
        printf("%s", input);
        first = 0;

        int val = atoi(input);

        if (val == 999) {
            printf(". ");
            break;
        }

        if (val >= 0 && val <= 12) {
            for (int i = 0; nums[val][i] != '\0'; i++) {
                letter_counts[nums[val][i] - 'A']++;
            }
        }
    }

    int n;
    for(n=0;n<26;n++) {
        for(int j=0; j<letter_counts[n]; j++) {
            printf("%c", 'A' + n);
        }
    }
    printf("\n");
    return 0;
}
```

Output:

```
2 3 0999
2 3 0999. EEHORTTW
```

```
8 14 0999
8 14 0999. EGHIT
```